6,906: Economics: Macro for Computer Science

Lecture 2: Key Concepts in Macroeconomics

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Learning objectives

- By the end of today's lecture, you should be able to:
  - Analyze the three central variables in macroeconomics: output, unemployment, and inflation;
  - Calculate and interpret output growth, unemployment rate, and inflation rate;
  - Describe the relationship between the three variables over time;
  - Differentiate between the business cycle and long-run approach to macroeconomics;

Output

Aggregate output

- The measure of aggregate output is called gross domestic product (GDP) and is usually denoted by Y.
- How would you define aggregate output in the economy?
  - **Aggregate production**:
    - GDP is the value of **final** goods and services produced in an economy during a given period. Note that we count only final goods, not intermediate goods.
    - GDP is the sum of **value added** in the economy during a given period, where the value added by a firm is the value of its production minus the value of the intermediate goods used.
  - **Aggregate income**:
    - GDP is the sum of all payments to the factors used in production (labor, capital) during a given period.
  - Aggregate production and aggregate income are always equal.

Aggregate output

- Consider an economy with two firms, Firm 1 and Firm 2. Is aggregate output the sum of the values of all goods produced, i.e., $300? Or just the value of cars, i.e., $200 dollars?

Steel Company (Firm 1)		
Revenues from sales		$100
Expenses		$80
Wages	$80	
Profit		$20

Car Company (Firm 2)		
Revenues from sales		$200
Expenses		$170
Wages	$70	
Steel purchases	$100	
Profit		$30

Source: Blanchard (2021)

Aggregate output

- Aggregate income is spent on consumption (C), investment (I), government expenditure (G), and net exports (NX), which are defined as the difference between exports (X) and imports (IM), where exports are defined as goods and services produced domestically and sold abroad and imports are defined as goods and services produced abroad and purchased domestically.

$$Y_t = C_t + I_t + G_t + NX_t = C_t + I_t + G_t + (X_t - IM_t)$$

- This is based on the assumption that the aggregate economy is populated by four actors: households, firms, government, and the rest of the world.
- Hence, this relationship indicates that aggregate expenditure, the sum of all spending on final goods produced within a country, equals aggregate income, which in turn equals aggregate production.

Aggregate output

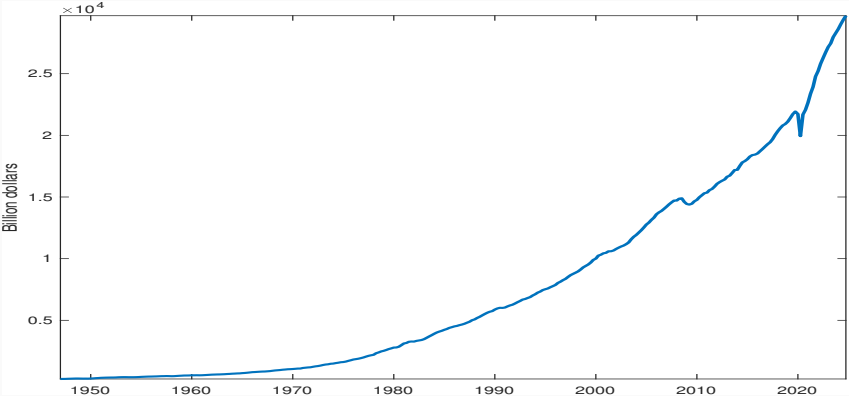
- Why do we subtract imports if GDP is a measure of production in a country in a given period, while imports measure production from other countries?

Aggregate output

- Why do we subtract imports if GDP is a measure of production in a country in a given period, while imports measure production from other countries?
- The reason is that GDP is the value of goods and services produced within a country, but expenditure on consumption, investment, and government spending do not distinguish between goods and services produced domestically or abroad.

Aggregate output

- Let's look at the plot of quarterly GDP, in billions of dollars, seasonally adjusted for the United States from 1947Q1 to 2024Q4.
- How would you interpret the evolution of this series?



Source: Federal Reserve Economic Data (FRED)

Nominal versus real GDP

- This measure of GDP is called nominal GDP and is measured in current prices.
- Nominal GDP is the sum of the quantities of final goods produced times their current price.
- Nominal GDP changes over time for two reasons:
  - The quantities of goods change;
  - The price of goods changes.
- Our goal is to measure production (equivalent to income and expenditure) and its change over time.
- Real GDP is defined as the sum of quantities of final goods times constant (not current) prices.
- Constant (real) prices are denominated in units of goods, whereas nominal prices are denominated in units of money.

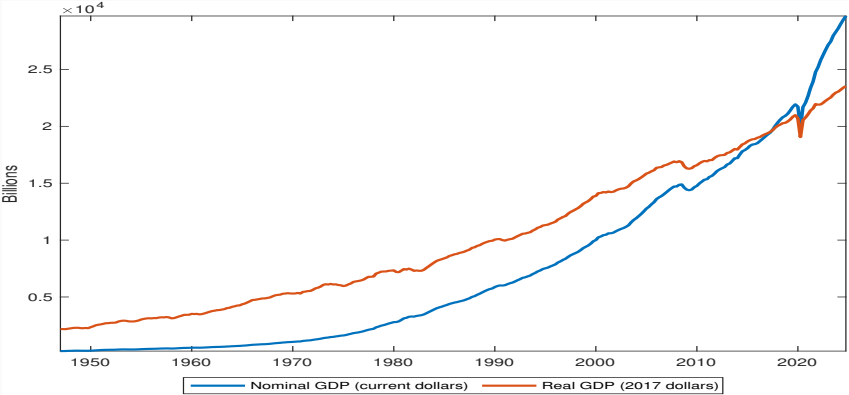
Nominal versus real GDP

Year	Quantity of Cars	Price of Cars	Nominal GDP	Real GDP (in 2012 dollars)
2011	10	$20,000	$200,000	$240,000
2012	12	$24,000	$288,000	$288,000
2013	13	$26,000	$338,000	$312,000

Source: Blanchard (2021)

Nominal versus real GDP

- Real GDP in chained (2017) dollars reflects relative prices that change over time. The year used to construct prices is called the base year. Note that the choice of the base year is arbitrary.



Source: Federal Reserve Economic Data (FRED)

Nominal versus real GDP

For the U.S.,

- Nominal GDP is also called dollar GDP, or GDP in current dollars.
- Real GDP is also called GDP in terms of goods, GDP in constant dollars, GDP adjusted for inflation, GDP in chained (2017) dollars, or GDP in 2017 dollars.

In our class,

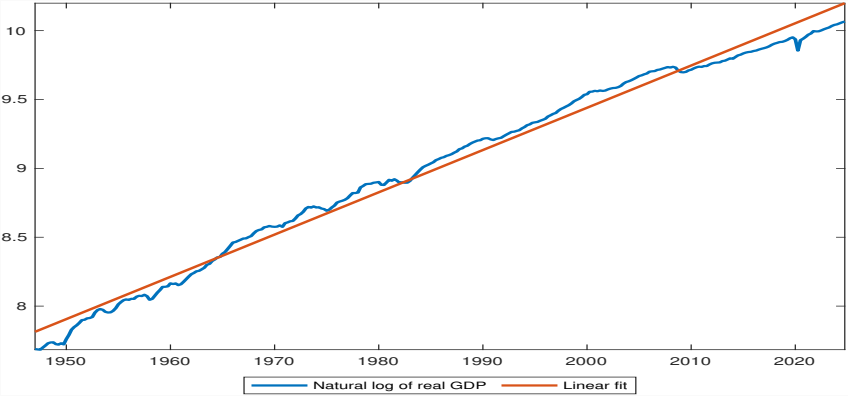
- We will use GDP to refer to real GDP. Y_t will denote real GDP in year t.
- Nominal GDP and variables in current dollars will be denoted by a dollar sign in front of them, e.g., $\$Y_t$.

Natural logarithm of real GDP

- GDP grows exponentially over time which implies the slope gets steeper as time goes by.
- The natural log of an exponential function is a linear function that is much easier to interpret.
- The slope of a plot in the log is approximately just the growth rate of the series.
- This means that we can interpret differences in the log across time as (approximately) percentage differences.

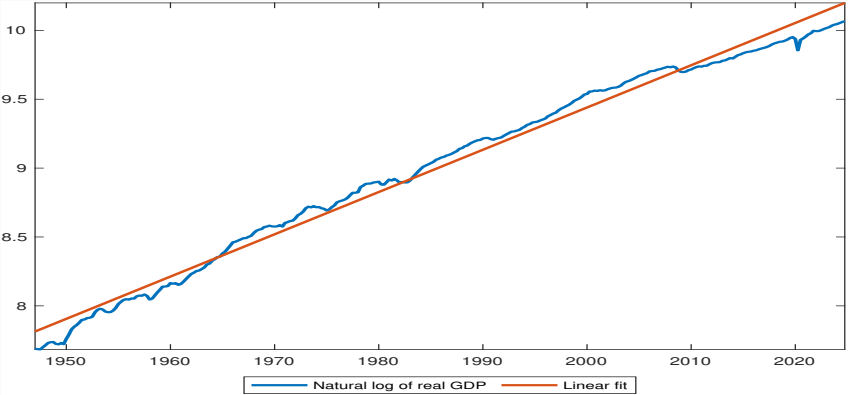
$$ln(Y_t) - ln(Y_{t-1}) \approx \frac{Y_t - Y_{t-1}}{Y_{t-1}}$$

Natural logarithm of real GDP



Source: Federal Reserve Economic Data (FRED)

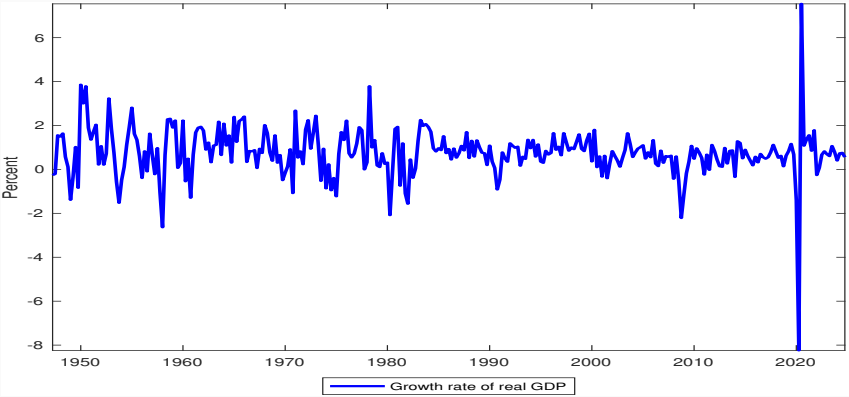
Natural logarithm of real GDP



Source: Federal Reserve Economic Data (FRED)

Note that a straight line can approximate log real GDP over time pretty well. Since we are looking at the natural log, this means roughly constant trend growth over time.

Growth of real GDP



Source: Federal Reserve Economic Data (FRED)

The average growth rate (log first difference) of quarterly nominal GDP from 1947-2024 was about 0.0076, or 0.76 percent. This translates into an annualized rate (what is most often reported) of about 3.04 percent (approximately 0.76 x 4).

Growth of real GDP

- If the real GDP growth rate is about 3.04 percent per year, would you expect the growth rate of nominal GDP to be higher or lower?

Growth of real GDP

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  - The growth rate of nominal GDP over the same period is 6.18 percent.

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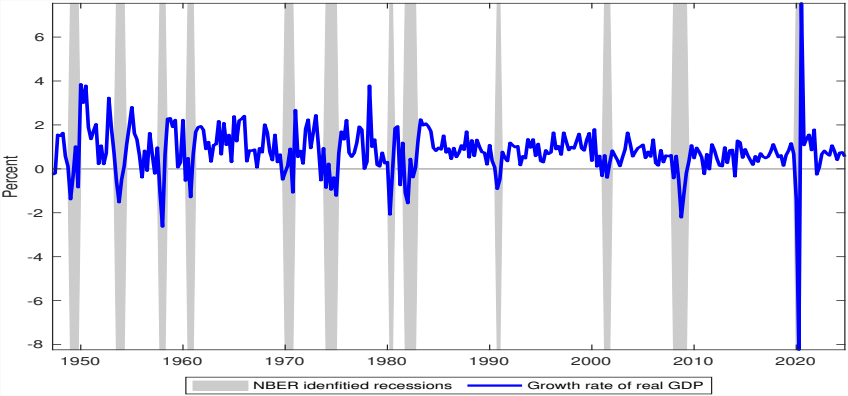
  - The growth in nominal GDP should approximately equal growth in prices (inflation) plus growth in real GDP.
- Could you use the real GDP growth rate to identify recessions?
  - There is no formal definition of a recession, but loosely speaking a recession is defined as two or more quarters of a sustained slowdown in overall economic activity.

Growth of real GDP



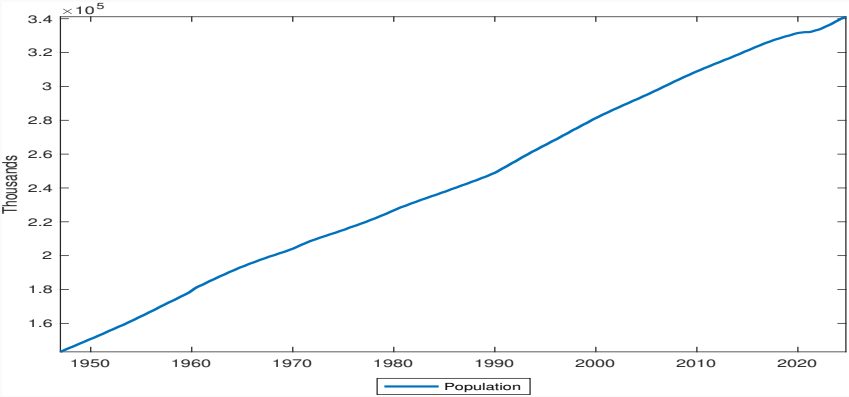
Source: Federal Reserve Economic Data (FRED)

- By eyeballing the graph, we see GDP declining in most recessions.

Real GDP per capita

- Real GDP can go up either because more people are working in an economy or because the people working in an economy are producing more.
- Hence, we need to consider that at least a part of the increase in real GDP over time is due to population growth.
- For thinking about an economy's productive capacity, and for making comparisons across time, we want a measure of GDP that controls for the number of people working in an economy.
- For this reason, it is common to look at "per capita" measures, which are series divided by the total population.

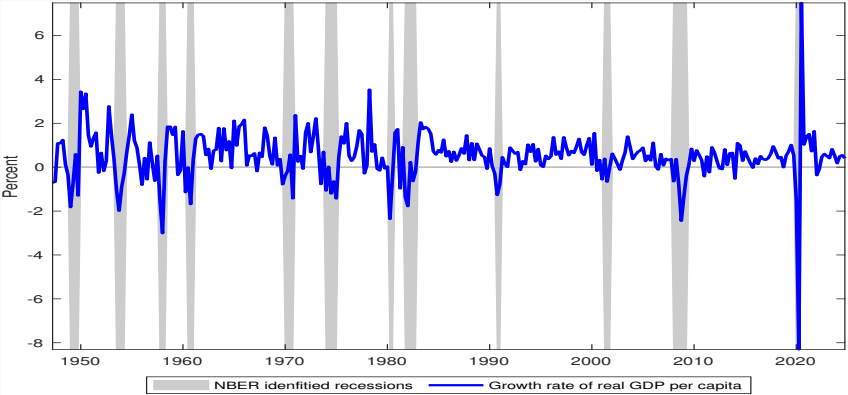
Population



Source: Federal Reserve Economic Data (FRED)

- In the U.S., population growth has been pretty smooth over time. Since the end of WW2, it has averaged about 0.28 percent per quarter, which translates to about 1.12 percent per year.

Real GDP per capita

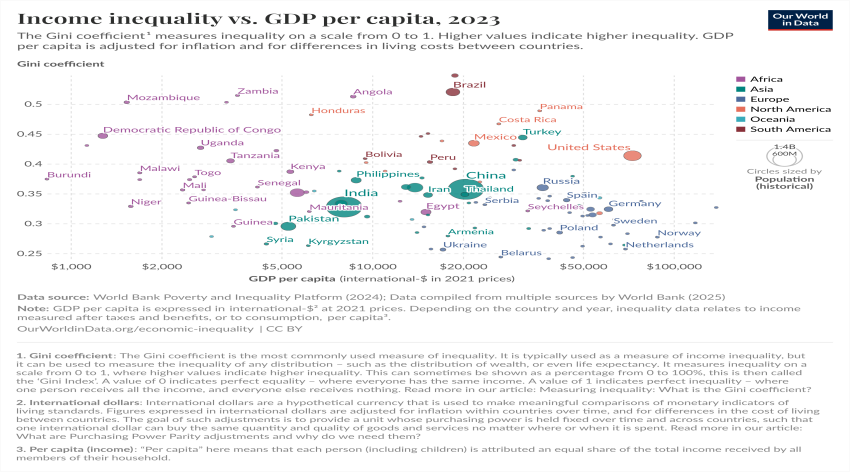


Source: Federal Reserve Economic Data (FRED)

- Across time, the average growth rate of real GDP per capita has been 0.49 percent per quarter or close to 2 percent per year.

Real GDP per capita as a measure of economic health

- Is GDP per capita a good measure of an economy's prosperity?



Real GDP per capita as a measure of economic health

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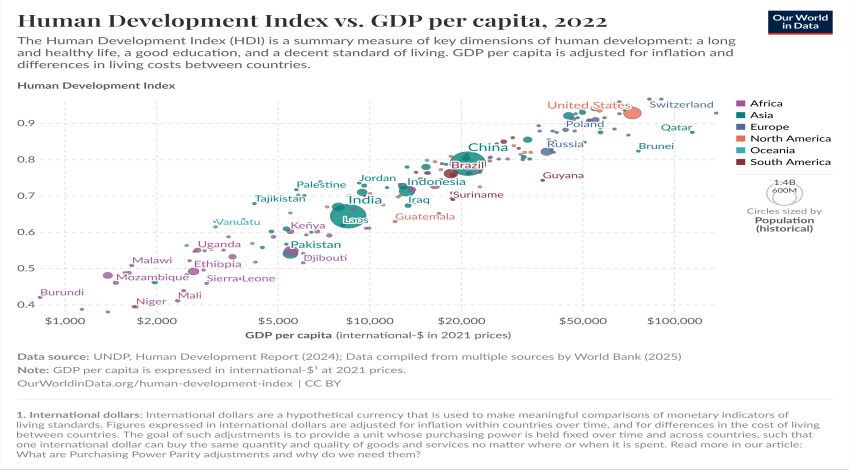


Unemployment

Unemployment rate

- How to get data on the number of unemployed?
- Because unemployment insurance goes only to people who have applied for such benefits and it is impractical to count every unemployed person each month, most rich countries rely on large surveys of households to compute the unemployment rate.
- Each month, the U.S. Bureau of Labor Statistics (BLS) announces the total number of employed and unemployed people in the U.S. for the previous month, along with many characteristics about them.
- The Current Population Survey (CPS) has been conducted in the U.S. every month since 1940. There are about 60,000 eligible households in the sample (about 110,000 individuals) each month. The CPS sample is selected to be representative of the entire civilian noninstitutional population of the U.S.
- The survey excludes people living in institutions (for example, a correctional institution or residential nursing or mental health care facility) and those on active duty in the Armed Forces.

Unemployment rate

- The basic concepts involved in identifying the employed and unemployed are quite simple:
- People with jobs (part-time and temporary work, as well as regular full-time, year-round employment) are employed.
- People are classified as unemployed if they do not have a job, have actively looked for work recently, and are currently available for work.
- The labor force comprises the employed and the unemployed, $L = E + U$.
- Those who have no job and are not looking for one are counted as not in the labor force.
- The unemployment rate is the ratio of the number of people who are unemployed to the number of people in the labor force, $u = U/L$.

Unemployment rate

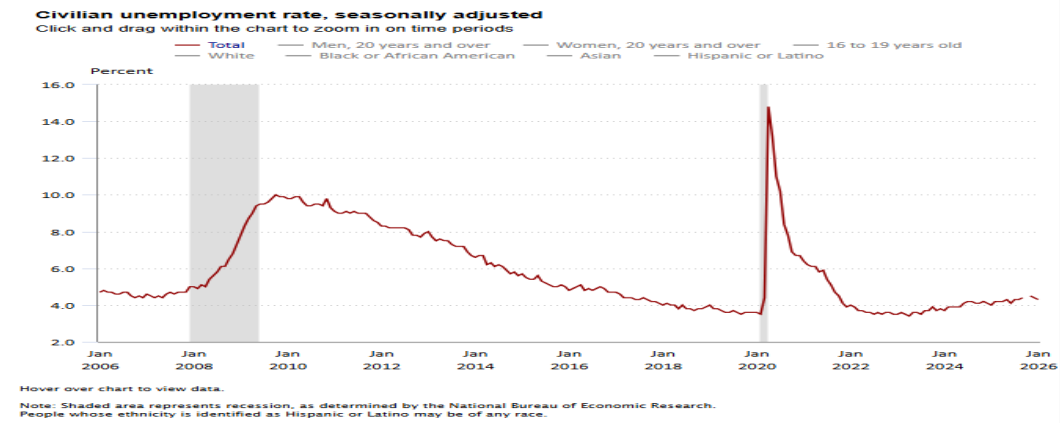
- The unemployment rate does not account for discouraged workers. They are those persons who give up looking for a job.
- The participation rate is the ratio of the labor force to the total population of working age.
- Because of discouraged workers, a higher unemployment rate is typically associated with a lower participation rate.
- The unemployment rate can move because either the number of unemployed changes or the number of employed changes. For example, the number of unemployed could fall if some who were officially unemployed quit looking for work, and are therefore counted as leaving the labor force, without any change in employment.

Unemployment rate

- Why do economists care about unemployment?
  - Because of its direct effect on the welfare of the unemployed, especially those remaining unemployed for long periods. Even if there is unemployment insurance in most countries, it does not insure a person's income fully and it covers only a limited period. Longer periods of involuntary unemployment may inflict serious psychological costs on the jobless, hence unemployment gives rise to serious individual problems.
  - At the level of society, unemployment is a social waste. It is a signal that the economy is not using its human resources efficiently. The income that the unemployed would have earned, had they had jobs, would have been taxed and the tax revenue used to benefit other citizens.
  - Note that very low unemployment can also be a problem as the economy runs into labor shortages.

Unemployment rate

- How would you explain the evolution of the unemployment rate in the U.S. over the period Jan. 2006- Jan. 2026?

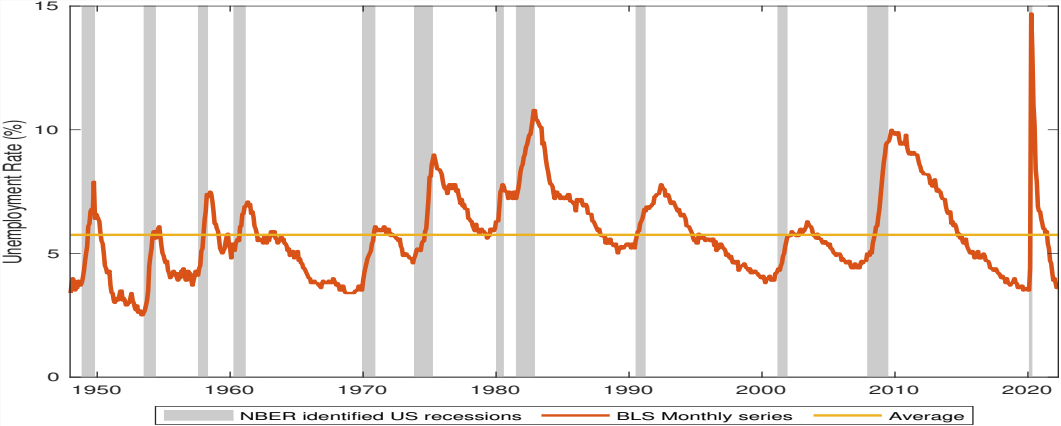


Unemployment rate

- There are short-run (cyclical) fluctuations of unemployment around the average rate of unemployment (natural rate of unemployment), sometimes large (14.7 % in April 2020)!
- These are mainly the patterns of cyclical unemployment: temporary changes in unemployment due to economic expansions or recessions.
- Why does unemployment fluctuate with the cycle?
- Nominal wage rigidity! This means that it takes time for nominal wages to adjust to economic circumstances.
- However, the spike in unemployment in 2020 is due to the public health response to the coronavirus pandemic, an economic shock itself, and is not the usual cyclical unemployment associated with a recession.

Unemployment rate

- The average unemployment rate over the period Jan. 1948 - Jan. 2026 is 5.7%.



Unemployment rate

- The average unemployment rate is positive in the long run. Why not zero?
- Real wage rigidity! This means there is full flexibility of prices and wages but an excess supply of labor still prevails.
- How can we have an unemployment rate of 5-7% and no downward pressure on wages?
  - Frictional unemployment: it takes time to move from one job to another.
  - Search unemployment: there is uncertainty about job possibilities, and therefore the unemployed choose to remain unemployed for some time while searching for jobs.
  - Efficiency wages: the labor market is competitive, but labor productivity depends on wages.
  - Trade unions: the labor market is not competitive.

Trade unions

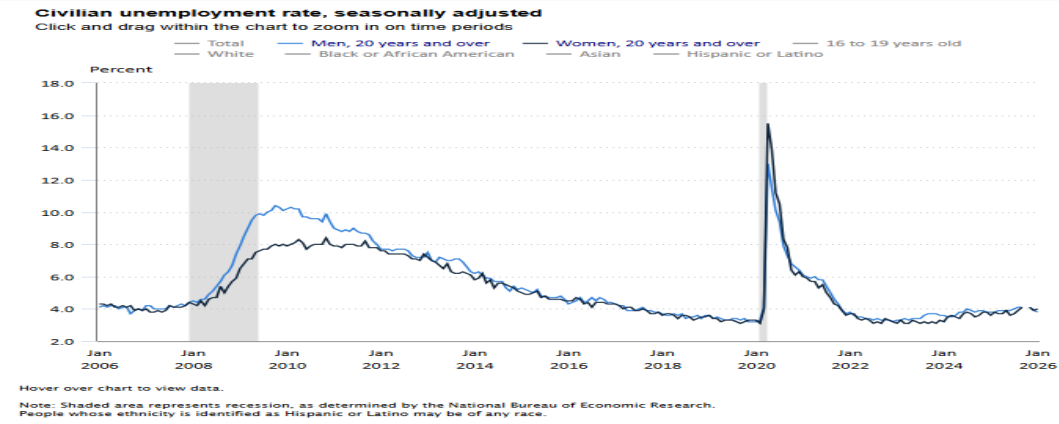
- Sometimes wages are set by collective bargaining-a bargaining between unions and firms.
- Slightly more than 10% of U.S. workers' wages are set by collective bargaining.
- The higher the skills needed to do the job, the more likely there is to be bargaining between employers and individual employees.
- Collective bargaining plays an important role in Japan and most European countries.

Efficiency wages

- Health argument (less important in developed countries): Better health means better productivity.
- Keeping the workers in the firm: When unemployment is low, firms that want to avoid an increase in quits and will increase wages to induce workers to stay with the firms.
- Getting good workers at hiring: By offering higher wages, firms may be able to recruit more skilled applicants.
- The shirking argument: Firms may want to pay a wage above the reservation wage (the wage that would make them indifferent between working or being unemployed) in order to decrease workers' turnover and increase productivity.
- Reciprocity: Treating others like they treat you.

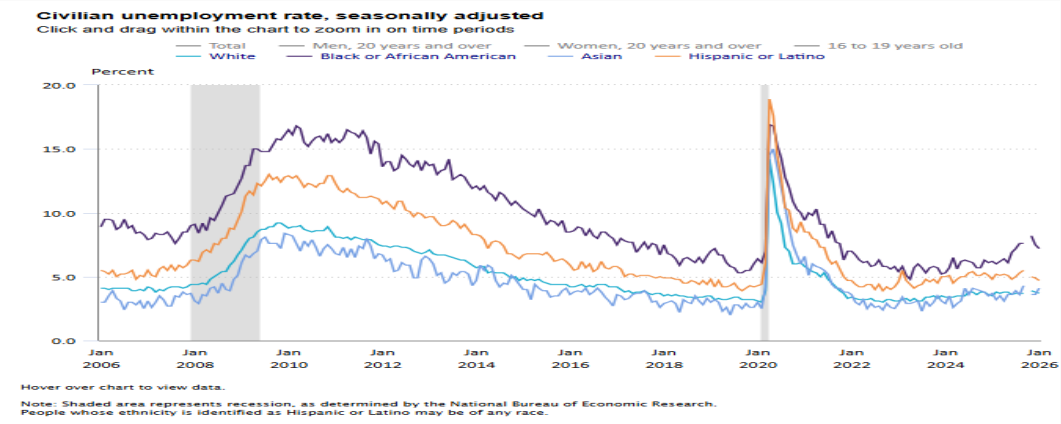
Heterogeneity in unemployment rates

- Variation in unemployment rates by gender:



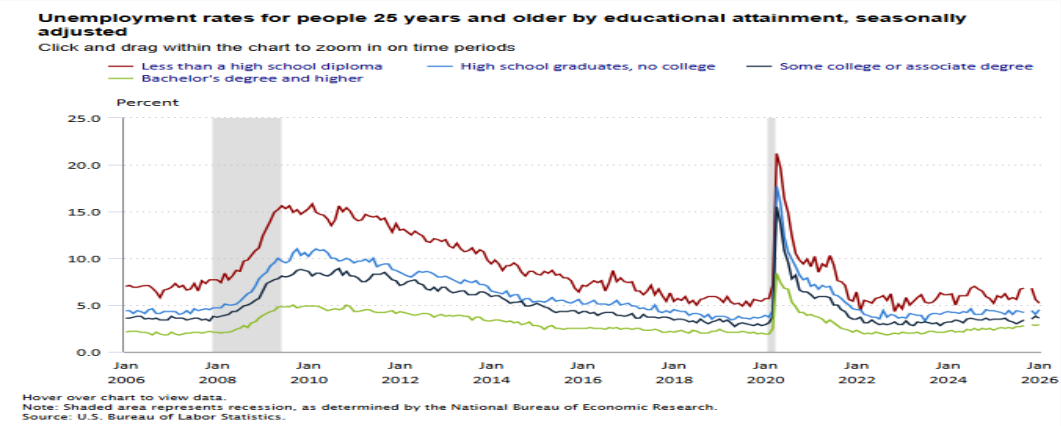
Heterogeneity in unemployment rates

- Variation in unemployment rates by ethnicity:

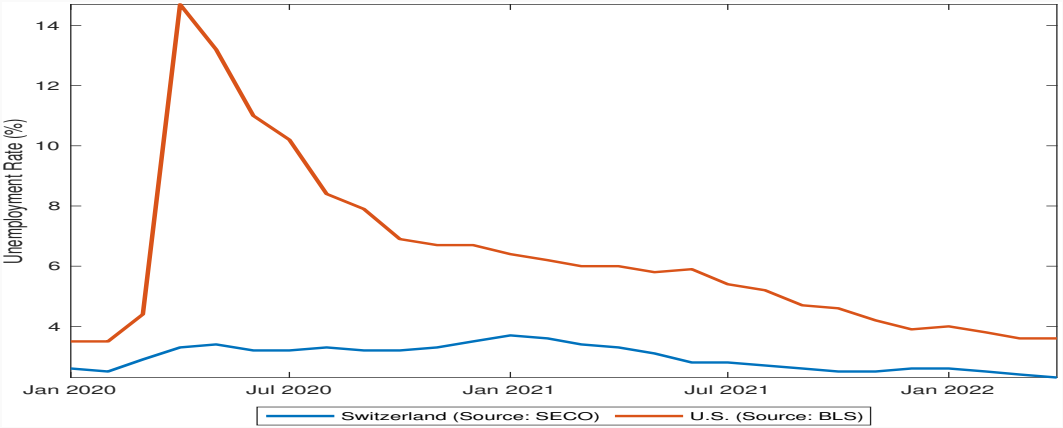


Heterogeneity in unemployment rates

- Variation in unemployment rates by educational attainment:



Comparison of unemployment rates: U.S. and Switzerland



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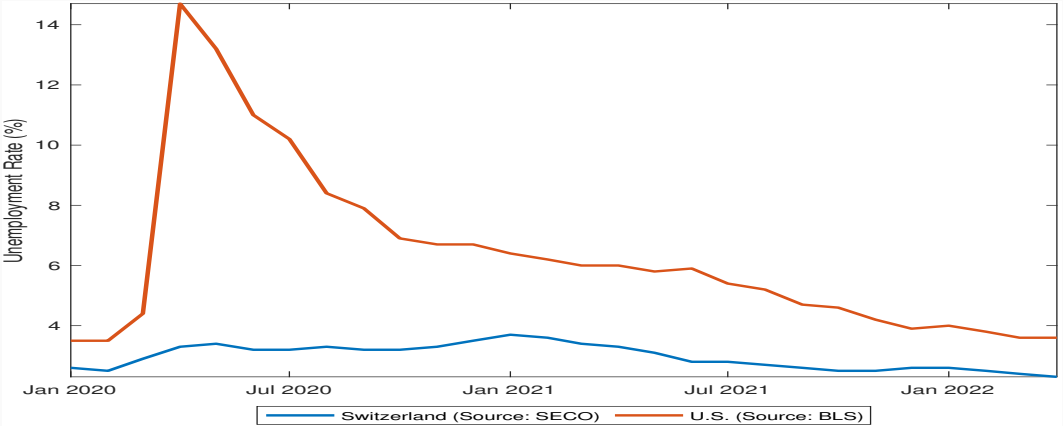


- The State Secretariat for Economic Affairs (SECO) unemployment rate does not count job seekers who do not receive unemployment benefits. The comparison is wrong!

Unemployment rate

	Switzerland (Source: SECO)	Switzerland (Source: ILO)
2020-01-01	2.6	4.4
2020-02-01	2.5	4.5
2020-03-01	2.9	4.6
2020-04-01	3.3	4.5
2020-05-01	3.4	4.5
2020-06-01	3.2	4.7
2020-07-01	3.2	5.2
2020-08-01	3.3	5.3
2020-09-01	3.2	5.2
2020-10-01	3.2	4.9
2020-11-01	3.3	4.9
2020-12-01	3.5	5.0

- An alternative measure, used by the International Labour Organisation (ILO), which includes all available job seekers and is based on survey (closer to BLS approach), is higher. In 2020, the SECO rate was 2.6% while the ILO rate was 4.8%.
- European governments often provide financial aid to firms to keep workers on the payroll, while U.S. policymakers rely mainly on unemployment insurance to support those losing their jobs.

Inflation

Inflation rate

- Inflation is a sustained rise in the general level of prices - the price level.
- The inflation rate is the rate at which the price level increases: $\Pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$
- Deflation is a sustained decline in the price level (negative inflation rate).
- Two prominent methods to compute the inflation rate - based on GDP deflator or based on the Consumer Price Index (CPI).

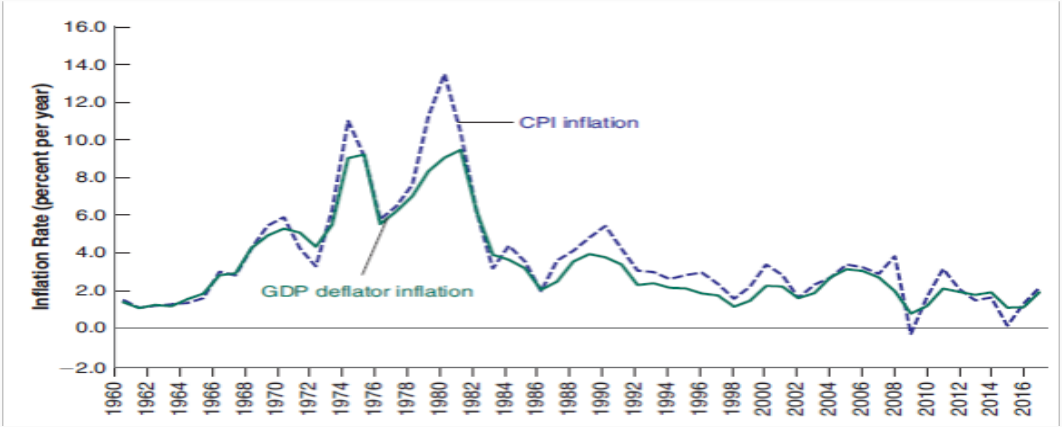
Inflation rate based on GDP deflator

- The GDP deflator in year t (P_t) is the ratio of nominal GDP to real GDP in year t: $P_t = \frac{Nominal\ GDP_t}{Real\ GDP_t} = \frac{\$Y_t}{Y_t}$.
- It is called an index number (e.g. 1 in 2012), which has no economic interpretation.
- The rate of change has a clear interpretation: the rate of inflation.
- The rate of growth of nominal GDP is equal to the rate of inflation plus the rate of growth of real GDP.

Inflation rate based on CPI

- The set of goods produced in the economy is not the same as the set of goods purchased by consumers because:
  - Some of the goods in GDP are sold not to consumers but to firms, to the government, or to foreigners.
  - Some of the goods bought by consumers are not produced domestically but are imported from abroad.
- The Consumer Price Index (CPI) is a measure of the cost of living.
- For the U.S., the CPI is published monthly by the Bureau of Labor Statistics (BLS), which collects price data for 211 items in 38 cities.
- The CPI gives the cost in dollars of a specific list of goods and services over time. The building block of the CPI is a "consumption basket." The basket includes both different kinds of products and different quantities. The basket may include 12 gallons of milk, 40 gallons of gasoline, 4 pounds of coffee, etc.

Comparison of inflation rates



Source: Blanchard (2021), based on data from Federal Reserve Economic Data (FRED)

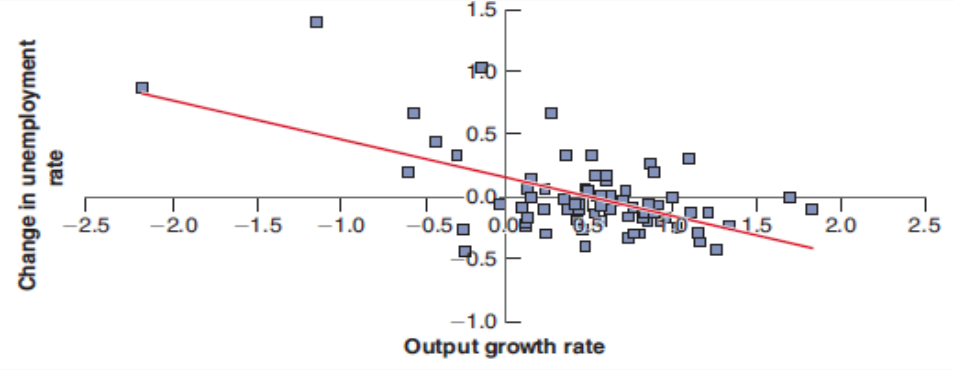
- The inflation rates, computed using either the CPI or the GDP deflator, are largely similar.

Comparison of inflation rates

- The CPI and GDP deflator moved together most of the time.
- Exception: In 1979 and 1980, the increase in the CPI was significantly larger than the increase in the GDP deflator due to the price of imported goods increasing relative to the price of domestically produced goods.
- Remember that the GDP deflator is the price of goods produced in the U.S., whereas the CPI is the price of goods consumed in the U.S.
- For getting a sense of overall price inflation in US-produced goods, the GDP deflator is preferred.
- For getting a sense of nominal changes in the cost of living for the average household, the CPI is a good measure.

Output, unemployment, and inflation

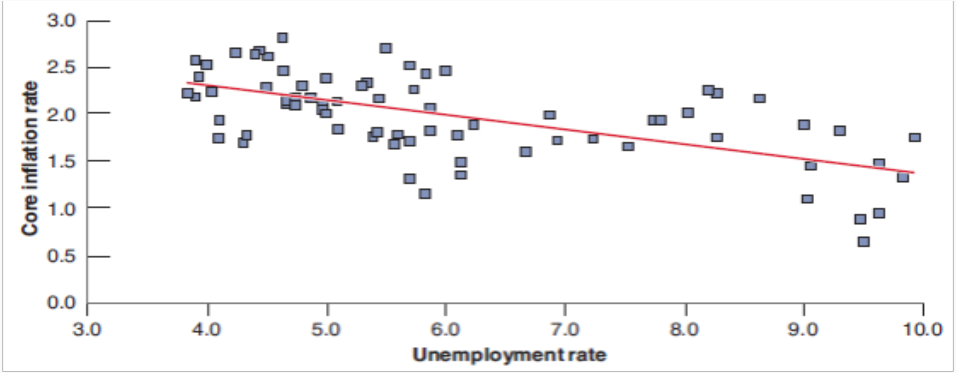
Okun's Law



Source: Blanchard (2021), based on data from Federal Reserve Economic Data (FRED)

- Output growth that is higher than usual is associated with a reduction in the unemployment rate.
- Output growth that is lower than usual is associated with an increase in the unemployment rate.

Phillips Curve



Source: Blanchard (2021), based on data from Federal Reserve Economic Data (FRED)

- A low unemployment rate is associated with an increase in the inflation rate.
- A high unemployment rate is associated with a decrease in the inflation rate.

Summary

- In this lecture we learned:
  - Calculate and interpret output growth, unemployment rate, and inflation rate
  - To differentiate between nominal and real variables.
  - To describe the relationship between output growth, unemployment rate, and inflation rate.
  - To differentiate between the business cycle and long-run study of macroeconomics.
- Next lecture we will study one of the workhorse macroeconomic models.